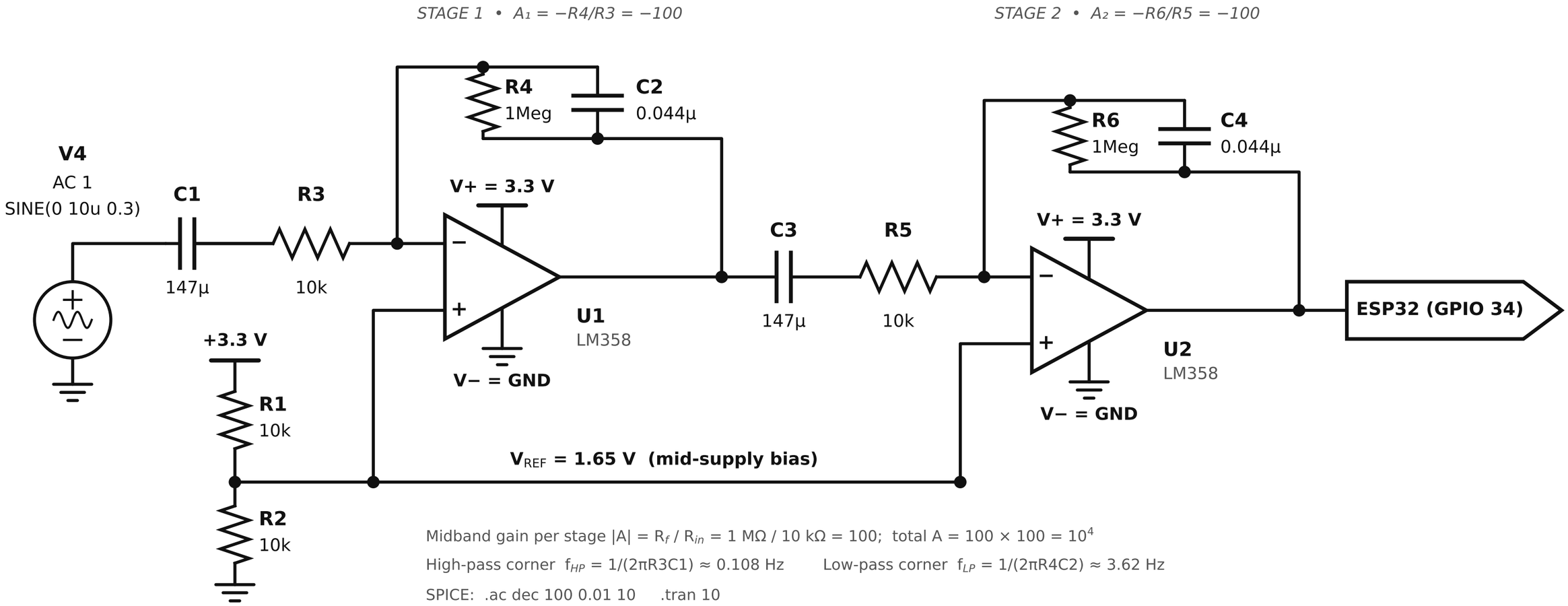
MachineSense: Non-Contact RF Vital Sign Monitor

Analog front-end: two-stage inverting LM358 band-pass amplifier, total gain 10,000 (single 3.3 V supply)



Midband gain per stage $|A| = R_f / R_{in} = 1\ M\Omega / 10\ k\Omega = 100$; total $A = 100 \times 100 = 10^4$

High-pass corner $f_{HP} = 1/(2\pi R3C1) \approx 0.108$ Hz Low-pass corner $f_{LP} = 1/(2\pi R4C2) \approx 3.62$ Hz

SPICE: .ac dec 100 0.01 10 .tran 10